2. Data Warehouse :

- This is a central storage for large historical structured data from different sources specifically for analizing and Reporting.

- Data warehouse Helps stores data for analitical processing while Data base is for day-day updates.

- Data Warehouse is ~~for~~ to help understand and Improve Business over time where a company uses it to Make Informed business decisions, Identify trends and patterns and Also Assume future outcomes.

3. Data Lake

- This is a centralised Location where Raw data is kept in its original format, It is Highly Flexible and designed to Handle Big data from Many Sources.

- Data Lake stores big amounts of Raw, unstructured | structured ~~for~~ Data and is Highly Flexible, Data Warehouse on the other hand stores primarily structured data and is ~~for~~ ~~Highly~~ used as curated repository for reporting.

- A Business May choose Data Lake over data Warehouse because it is Cost Effective, lower Storage Costs & is highly flexible.

4. Data Mart

- This is a curated database that is designed to serve the specific needs of a single data team, Data Mart make specific data available to a defined Group of users which allows those users to quickly Access Information.

- Data Mart is Focused version of data warehouse that Contains specific Information within a Data Warehouse.

5. Structured Data:

- Highly Organized Information Formatted Into Rows and Columns making It easiH searchable, Analyzable and Readable.
- i.e. Music Libraries.

② Semi - Structured data.

- This Is the category of data that does not follow Rigid structure, It Is the bridge between Structured & unstructured data.
- i.e An Email

③ UNSTRUCTURED Data.

- This Is data that has No predefined Format or organization, It does Not fit neatly Into tables and cannot be Stored or analyzed.
i.e. REVIEWS on Site by a User.

6. Data Modelling;

- Process of creating Visual diagrams that define How data Is Structured, Stored, and used within an organization.
- It Is Important because It Creates Structural blueprint Ensuring data accuracy, Reducing errors, It acts As a bridge business Requirements and technical Implementation.

- i.e @ E-commerce Stores With their clear blueprint, Customer-centric design.

7. What is Data Mining? How is it used in business? provide two real-world examples of data mining in action.

- Data mining is the process of analyzing massive datasets to discover hidden patterns, anomalies, correlations and trends that predict future outcomes

The use of Data mining

- Data mining in business transforms large datasets into actionable insights, driving smarter decisions through predictive analytics, customer profiling, and trend analysis.

Real-word Examples.

- Walmart - Famously analyzed their transactions data and discovered that before a hurricane, sales of pop Tarts increased significally, alongside flash light sales.

Banking / Finance

- Banks use data mining to analyze massive volumes of financial transac in real time to identify abnormal behaviour.
- If a credit card shows a sudden high value purchase in a foreign country, a pattern that deviates from the users typical behaviour, the da mining algorithm flags it instantly.

8. What is ETL (extract

8. What is a Database Management System (DBMS) Name and briefly d four popular DBMS platforms (eg, mysql, postgresql, Oracle, MongoDB and Explain a suitable use case for each.

- Database management system is a software that acts as an interf between users, applications and data, allowing for the efficient creation, re updating and management of structured or unstructured data.

Four popular database management systems.

1. MySQL - An Open-Source Relational Database management System that uses structured Query language (SQL). It is renowned for it's speed, reliability and ease for use, making it the standard choice for web applications, Content management Systems (like wordpress), and Small to-medium Sized projects.

2. postgreSQL
- A powerful, Open-Source object-relational database System known for it's advanced features, high SQL Compliance, and extensibility. It is often preferred for Complex, Mission Critical applications that require data integrity and Support for both relational (SQL) and non-relational (Json querying.

3. MongoDB
- A leading Open-Source NoSQL document-Oriented database. Unlike traditional RDBMS, it uses flexible, Json-like documents ~~with~~ dynamic ~~Sthem~~ rather than tables. It is highly popular for it's Schema-Free approach, allowing developers to easily manage rapidly changing, Unstructural data making it ideal for big data, Content management and mobile apps.

4. Oracle Database
- A highly robust, Commercial and multi-model RDBMS developed by Oracle Corporation. It is a robust System geared toward large enterprises for mission-Critical applications, data warehousing and Complex transaction processing.

9) Extract Transform load stand, it is a process used to collect data from different sources, prepare it, and store it for analysis

Extract

This is the first step where data is collected from various source such as databases, spreadsheets, websites, or applications

Example

Pulling sales data from a company's system

Transform

In this step, the extracted data is cleaned and organized into a useful format. This may include

- Removing errors or duplicates
- converting data type
- combining data
- Apply calculation

Load

The final step is loading the clean data into a storage system like a data warehouse or databases where it can be analyzed

Example: saving processed data into a reporting system

ETL is important in data analytics
- Ensure data is accurate and consistent
- Makes data easier to analyze and understand
- combines data from multiple sources into one place
- saves time by automating data ~~pre~~ position preparation
- Help organization make better decisions based on reliable data

(10) • Comma separated values
• A simple text format where data is stored in rows and columns
• Easy to read and open in tools like EXCEL

when To USE:
• Small datasets
• Simple data Exchange Between Systems
• when human Readability is important

(2) JSON (Javascript object Notation)
• Stores data in a structured, key - value format
• Can handle Nested data (data within data)

when To USE:
• web Application And APIs
• when working with Complex or hierarchical data

(11) Artificial Intelligence (AI) is a field of Computer Science dedicated to Creating Systems Capable of Performing Tasks that Typically Require human Intelligence.
Narrow AI (weak AI) is designed To Perform Specific Task within a limited Context focusing on Efficiency In a Single domain. General AI (Strong AI/AGI) is a theoretical form that can learn, Understand, and apply intelligence To any Task a human can Possessing Self-awareness and Human like Cognitive Versality

(12) Machine learning (ML) is a Subset of artificial Intelligence (AI) that Enable Computer To learn from data and improve at Task without being Explicity Programmed for Every Rule

Three Types of Machine learning
- Supervised learning
- Unsupervised learning
- Reinforcement learning

(13) Deep learning a subset of machine learning driven by multi-layered artificial neural networks inspired by human brain

Machine learning (ML) is a subset of Artificial intelligence (AI), acting as the practical toolkit algarithms and statistical models that enables computer to learn from data, identify patterns, and make decisions without explict programming

A type of machine learning model designed to simulate the human brain's interconnected neuron structure, allowing computer to recognize patterns, analyze complex data and improve performance over time

14) Natural language Processing (NLP) is a field of AI that enables computers to understand, interpret, and respond to human language spoken or written

Tree real world applications

1. Chatbots and virtual assistants
Example - whatsapp chatbots or google assistant

2. Translation tools
example: Google Translate

15. What is a large language model (LLM)? How is it different from a standard machine learning model? Name well-known LLMs and explain what they used for.

Answer - A large ~~AI~~ language model (LLM) is a type of AI program designed to understand, process and generate human-like text by analyzing massive datasets. Based on deep learning and transformer models they predict subsequent text in a sequence, allowing them to answer questions, write code, translate languages and summarize documents

Differences

Standard machine learning is a ~~subfield of artif~~ is a category of machine learning where algorithm is trained on a dataset that includes input and their corresponding known outputs. While large language model can generate human-like text, conduct conversations, and summarize content. In exchange, they require vast and diverse datasets, huge ~~computers~~ compute resources, and careful fine-tuning to align output with intent.

Well-known LLMs

Are the AI systems trained on vast amounts of text data to understand, generate, and process human language and other structured data, such as computer code. They are built on transformer architectures that allow them to handle complex unstructured language at scale.

16. What is Generative AI? How does it work at high level? Name three tools that use Generative AI and describe what each one does.

- Generative AI is a subfield of artificial intelligence that uses generative models to generate text, images, videos, audio, software code or other forms of data. It works at a high level by utilizing deep learning models - specifically artificial neural networks - to analyze vast amounts of data, learn underlying patterns, and then create new original content

that mirrors human - like expression.

The three

- ChatGPT - A Conversational AI developed by Open AI that can produce human - like text, summarize documents, write code and and conduct web research.

- Elevenlabs - A platform specializing in high quality text - to - speech, voice synthesis, and voice cloning, often used to create natural ~~sure~~ sounding AI voiceovers.

- Midjourney - is an AI - powered platform that generates high - quality, detailed images from natural language text descriptions (known as prompts).

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect.

- Training data set is information that is used to teach a machine learning model how to make predictions, recognize patterns or generate content.

The important of quality of training when building

- The quality of training data is important when building an AI model because the model's performance, accuracy and reliability are directly determined by the data it "learns" from. Poor quality data leads to the "garbage in, garbage out" phenomenon, when flawed, biased, or incomplete inputs results in unreliable outputs, regardless of how sophisticated the model's algorithm is.

18. What is Computer Vision? How do machines 'see' and interpret images provide two real - word use cases of Computer Vision.

- Computer vision is a field of artificial intelligence (AI) that ~~makes~~ enables computers and systems to derive meaningful information from digital images, videos and other visual input allowing them to take action or make recommendations.

Machines interpreting images through a field of artificial intelligence (AI) known as computer vision, which processes visual data as numerical grids rather than as visual scenes.

Two Real-word of Computer vision

1. Healthcare : Medical image Diagnosis (Radiology).

- Computer vision algorithms are trained on vast datasets of medical images - X-rays, MRIs and CT Scans - to assist radiologists in detecting abnormalities.

2. Manufacturing : Automated Defect Detection.

- Factories use high-speed cameras integrated with deep learning models (specifically Convolutional Neural Networks) to inspect product on assembly lines in real-time.

1a. APPLICATION PROGRAMMING INTERFACE

- API is a way for different software systems to communicate with each other.
- Like in a Restaurant, you either tell the waiter what you want, the waiter takes the Request to the kitchen (system), and then brings back your food.

- API is Important because they allow different systems to share data.
- It also saves time, No need to build anything from scratch.
- Enables Intergrations e.g, Apps using payment gateways, Maps.
- Power Modern apps like mobile apps, Websites, and Cloud Sezvices.

2a. AN ~~A~~ API key.

- THIS is a Unique code used to identify and auntheticate a user or application accessing API.

- Purpose of API is to Control ACCESS to API, track usage and help prevent misuse.

- How should API be stored and protected?
keep it a secret (Never share publicly)
- Store in Environment variables or secure severs.
- Do Not hardcode it in public code.

If API is EXPOSED PUBLICLY:
- Others can use without permission.
- You May be charged for their usage
- Your data or system could be compromised.

21. Rest API

- Representational State transfer is a type of API that uses standard web methods to communicate over HTTP.

Main HTTP METHODS:

GET - Retrieve data e.g. Get user info.

POST - Create new data e.g. Add new user.

PUT - Update existing data.

Delete - Remove Data.

22. JSON (JavaScript Object Notation)

- This is a lightweight format used to store and exchange data.

- Why is JSON commonly used?
- Easy to Read and Write.
- Works well with API's
- Supported by most programming languages

Example of JSON

```
{
  "name" : NOMTHANDA 20
  "age" : 22,
  "course" : "Data Analytics
  "isStudent" : true
```

23. PROMPT ENGINEERING

- THIS Is the Skill of Writing Clear and EFFective Instructio
for AI tools like ChatGPT to get better Results.

Why Is It Important?

- It Helps get accurate and Useful answers.
- SAVES time
- Improves AI Performance.

Example

- Poor Prompt : "EXPlain data"

- Improve Prompt :
"Explain What data Is In Simple terms and Give 3-real life
Examples.

24. What Is a Map Function.

- A MAP Function Is Used to apply a Function to Every Item In a List and Return a New List With the Results.

What It does ?

- Transforms Data
- PROCESSES Each element In a Collection

Python Example :

```
numbers = [1, 2, 3, 4]
Squared = list (map (lambda X : X ** 2, Numbers )
Print (squared)
```

Output

[1, 4, 9, 16]

25. Cloud Computing

- This is a delivery of computing services (like storage, servers, databases, and software) over the internet instead of using a local computer.

Main cloud service models.

- Iaas (Infrastructure as a service.
Provides basic computing resources like virtual machines and storage E.g.

Platform as a service (Paas)

- Provides a platform for developers to build and deploy applications without managing infrastructure.)
E.g:

Software as a service (SaaS)

- Provides software applications over the internet.
E.g:

26. Version control system (VCS)

- This is a tool that helps track changes to files and allows multiple people work on a project without overwriting each other's work.

Git

- This is a popular VCS used by developers and data analysts to manage code, track changes, and collaborate efficiently.

Why is is widely used?

- keep history of changes, allows teamwork and collaboration, makes it easy to fix mistakes.

Key TERMS :

- Commit : A Saved change or update In the Project.
- BRANCH : A SEPERATE Version OF a Project used to WORK on New Features Without affecting the Main Version.

27. Jupyter Notebook.

- This Is an Interactive tool that allows Users to Write and Run Code, View Results, and add Explanations all in One place.

- Most Commonly used Language Is Python

Why Is It popular :

- Combines Code, TEXT, and Visuals In One document.
- Easy For data analysis and Visualization
- Great For learning and Sharing WORK.
- Support Step-by-Step EXECUTION.

28. Python

- This Is a High-level programming Language known For It Simplicity and Readability.

- Why Is It popular In data analytics and AI ?

* Easy to learn and use
* Large Community Support
* Many powerful Libraries
* Work Well with data & machine learning.

Four common Python Libraries:

* Pandas: Used for data manipulation and analysis (table, datasets)
* Numpy: Used for numerical operations and working arrays
* Matplotlib: Used for creating graphs and charts.
* Scikit-learn: Used for machine learning and predictive modeling.

29. Data Privacy

- This refers to the protection of personal information and how it is collected, used, stored and shared.

Why is data privacy important:

- protects peoples personal information.
- prevents identity theft and fraud.
- Build trusts between users and organizations.

General Data protection Regulations (GDPR)

- This is the law that protects the personal data and privacy of individuals.

Where does it apply:

- It applies to any organization worldwide that processes data of E

30. ~~Wh~~ Data Ethics

- Data Ethics refers to the moral principles guiding how data is collected, used, and shared

THREE Examples OF UNETHICAL USE:

① Using personal data without Consent

Problem: Violetes PRIVACY and trusts.

* Biased AI decision-Making

PROBLEM: Can unfairly descriminate against Certein Groups.

* Selling User data Without permission.

- PROBLEM: Exploits Users For profit Without their knowledge.

31. Data Governance.

- This is a process OF Managing data availability, usability, Intergrity, and SECURITY Within an Organization.

key Components:

① Data EQUILITY - Ensuring data is accurate and Reliable.

② Data SECURITY - PROTECTING data from unauthenised access.

③ Policies & Standards - Rules for handling data.

④ Data Ownership - Assigning Responsibity for data.

⑤ Compliance - following Laws and Regulations.

32. BIAS In AI

- Happens en AI System produces unfair or prejudiced Results due to biased data or design

EXAmple:

- A Hiring System trained on past data may favor male candidates over female Candidates.

Consequences:

- Unfair treatment OF Individuals
- Reinforce discrimination
- Lost OF trust In AI systems.

33. Difference between Data Roles.

Role	Main Responsibility
Data Analyst	Analyzes data and creates reports.
Data Scientist	Builds models and predict trends.
Data Engineer	Builds data pipelines & systems.
AI Engineer	Develops AI models and systems.

34. Business Intelligence (BI)

- Refers to technologies and tools used to analyze data and help businesses make informed decisions.

Two popular BI tools:

1. Power BI
- Creates dashboards and reports
- Helps visualize business performance.

2. Tableau
- Turns data into interactive visuals
- Helps identify trends and patterns.

How BI helps businesses
- Supports better decision making.
- Identify trends and opportunities.
- Improves efficiency and performance.

35. This Assignment has shown be different types of Data Engineering, Data analysis and specifically LLMs, It Gave me an Insight of Some of the things I do daily/ use Everyday that I did Not Even pay Attention to, It opened my Eyes that all the data tools work together to form a proje

34. Data and ~~I~~AAI Aren't Just Abstract tech Ideas, they already Shaping how people live, work and make a decisions Everyday, by Giving personalized Experiences, Smart decision Making (In Banks use AI to detect Fraud.), Automation of tases (Chatbots, Voice Assistance) & better Service.

36. I Struggled With understandi Artificial Intelligence, Machine Learning and Natural Language processing.

③ Parquet

- A Column - based storage format (stores data by columns instead of rows)
- Highly compressed and efficient

When to use:

- Big data processing (e.g. large datasets)
- When performance and speed are important (e.g. analytics queries)

4. Avro

- A binary format that stores data along with its schema (structure)
- Supports data evolution (changing structure over time)

When to use:

- Data streaming systems
- When data structure may change over time
- Systems like big data pipelines (e.g. Apache Tools)

Summary

- Use CSV for simplicity
- Use JSON for flexible, nested data
- Use Parquet for fast analytics on large datasets
- Use Avro for scalable systems and evolving data structure

If you want, I can also help you turn this into a neatly formatted assignment or PDF

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